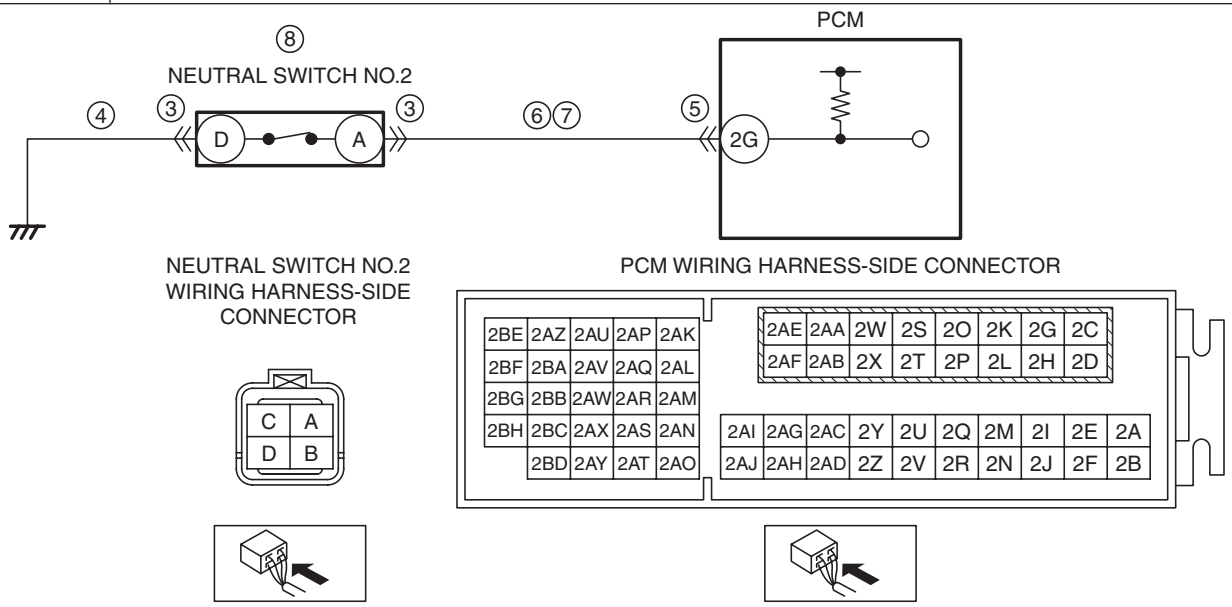
DTC P07BE:00 [SKYACTIV-D 1.5]

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DTC P07BE: 00	**Transmission indeterminate failure (failed to neutral)**
DETECTION CONDITION	• Correlation malfunction between neutral switches No.1 and No.2. **Diagnostic support note** • This is an intermittent monitor (other). • The check engine light does not illuminate. • FREEZE FRAME DATA/Snapshot data is not available. • DTC is not stored in the PCM memory.
FAIL-SAFE FUNCTION	Not applicable
POSSIBLE CAUSE	• Neutral switch No.2 connector or terminals malfunction • Open circuit in wiring harness between neutral switch No.2 terminal D and body ground • PCM connector or terminals malfunction • Short to ground in wiring harness between neutral switch No.2 terminal A and PCM terminal 2G • Open circuit in wiring harness between neutral switch No.2 terminal A and PCM terminal 2G • Neutral switch No.2 malfunction • PCM malfunction



Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
2	**VERIFY RELATED PENDING CODE AND/OR DTC** • Switch the ignition off, then ON (engine off). • Perform the Pending Trouble Code Access Procedure and DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the PENDING CODE/DTC P0850:00 also present?	Yes	Go to the applicable PENDING CODE or DTC inspection. (See DTC P0850:00 [SKYACTIV-D 1.5].)
		No	Go to the next step.
3	**INSPECT NEUTRAL SWITCH NO.2 CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the neutral switch No.2 connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 9.
		No	Go to the next step.

STEP	INSPECTION		ACTION
4	**INSPECT NEUTRAL SWITCH NO.2 GROUND CIRCUIT FOR OPEN CIRCUIT** • Verify that the neutral switch No.2 connector is disconnected. • Inspect for continuity between neutral switch No. 2 terminal D (wiring harness-side) and body ground. • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between neutral switch No.2 terminal D and body ground. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Inspect for the following: — Open circuit between neutral switch No.2 and body ground — Loose or lifting ground point • Repair or replace the malfunctioning part. Go to Step 9.
5	**INSPECT PCM CONNECTOR CONDITION** • Disconnect the PCM connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 9.
		No	Go to the next step.
6	**INSPECT NEUTRAL SWITCH NO.2 SIGNAL CIRCUIT FOR SHORT TO GROUND** • Verify that the neutral switch No.2 and PCM connectors are disconnected. • Inspect for continuity between neutral switch No. 2 terminal A (wiring harness-side) and body ground. • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between neutral switch No.2 terminal A and PCM terminal 2G. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. Go to Step 9.
		No	Go to the next step.
7	**INSPECT NEUTRAL SWITCH NO.2 SIGNAL CIRCUIT FOR OPEN CIRCUIT** • Verify that the neutral switch No.2 and PCM connectors are disconnected. • Inspect for continuity between neutral switch No. 2 terminal A (wiring harness-side) and PCM terminal 2G (wiring harness-side). • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between neutral switch No.2 terminal A and PCM terminal 2G. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 9.
8	**INSPECT NEUTRAL SWITCH NO.2** • Inspect the neutral switch No.2. (See NEUTRAL SWITCH INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the neutral switch No.2, then go to the next step. (See NEUTRAL SWITCH REMOVAL/INSTALLATION [F66M-R].)
		No	Go to the next step.

STEP	INSPECTION		ACTION
9	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Perform the DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.
10	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.